How Miyazaki "Discipline" His Players:

An Investigation of the Effects of Perceived Improvement on Players' Enjoyment

Coline Pickens, INFINITY, Sammy Elsayed, Sorceress Sellen*, Zhanshuo Dai

Library of Queen Rennala, Academy of Raya Lucaria Grand, Liurnia of the Lakes, The Lands Between

Abstract This study investigates the relationship between perceived improvement and players' enjoyment in a challenging platform game. We hypothesized that players who perceive improvement during gameplay would report higher enjoyment and stronger motivation to continue playing. To test this hypothesis, we developed a progressively difficult platform game and collected run-by-run survey data, gameplay performance data, and post-game survey responses. A Chi-square test was conducted to examine the association between perceived improvement and enjoyment levels. Results indicate a statistically significant relationship (x^2=28.961, p<.001), suggesting that perceived improvement positively influences player enjoyment. These findings contribute to the understanding of player motivation and provide practical implications for challenge-based game design.

Keywords Perceived player improvement, Game enjoyment, Player motivation

1. Introduction

The increasing popularity of the *Soulslike* genre, exemplified by games such as *Dark Souls*, has intensified academic and industry interest in the role of difficulty in player experience. Research consistently demonstrates that enjoyment emerges when game challenges align with player skill levels (Schmierbach et al., 2014; van den Hoogen et al., 2012).

While prior studies focus primarily on objective difficulty and challenge calibration, fewer studies have examined the psychological dimension of **perceived improvement**—that is, whether players believe they are getting better. Given that perceived competence is a central driver of intrinsic motivation, it is plausible that perceived improvement plays a critical role in enjoyment.

This study investigates whether players who perceive improvement between runs experience higher enjoyment and greater willingness to continue playing.

2. Literature Review

The enjoyment of video games is directly related to a player's skill and the challenge posed to them by the game. It has been well established that a player's enjoyment of a game is linked in part to how the challenge posed to them meets their skill (Schmierbach et al., van den Hoogen et al.). Players who were adequately challenged by the game relative to their skill resulted in the players receiving a greater level of enjoyment. Those who felt that the game was too easy or hard felt the opposite.

Furthermore, game enjoyment also hinges on how players perceive their skill (Klimmt et al.) and if they felt they improved (Alexander et al.). A player who thought they were better skilled enjoyed the game they played more than if they felt they were not very good regardless of how good a player actually was at the game. This links with the intersection between challenge and skill as adequately challenged players felt more skilled.

Additionally, different modes of difficulty scaling resulted in different benefits to player experience (Qin et al.). Different challenge experiences where challenge posed to players fluctuated had different levels of enjoyability based upon the length of the game players played. Difficulty scaling does not have a single best method, as different scalings are useful in different periods of time. To create more enjoyable experiences, difficulty should scale along with both a player's skill level, the amount of skill improvement they undergo, and the length of time that difficulty scaling occurs under.

3. Methodology

3.1 Game Design

In order to collect our data, we designed a simple platforming video game to test player improvement. The game consisted of one level that progressively got more difficult as it went on, and players would need to start from the beginning whenever they lost. It was expected that, as players played the game, they would get better at the game from run to run. The level was designed to be incredibly difficult near the end, so the players would be required to have improved a lot from their first run to the last. The movement mechanics were simple, but a mastery of them was similarly required to reach points in the game.

3.2 Data Collection

We collected three main sets of data. The first was gameplay footage. Each run was timed to track player improvement. If a player made it further in the game than any of their previous runs, they would be marked as having had a run that showed improvement.

The second set was a short survey that was done between runs. A player would mark on a sheet whether they felt that they improved or not and how much they enjoyed the previous run. Their improvement was a binary choice while their enjoyment was given on a five-point scale. The full survey instrument can be found in Appendix B.

The last piece of data that we collected was a post-run survey. Using this survey, we gauged how much they felt they improved, how much they felt they improved, and how much they would want to play the game again at a later date. The goal of this survey was to judge their overall experience outside of run-to-run mood fluctuation. The full survey instrument can be found in Appendix A.

The data we collected is intended to find a link between enjoyment, improvement, and perceived improvement. The run-to-run surveys are the main data set to meet this goal, but the post-game survey helps us gauge if their overall improvement was also a major factor.

4. Data Analysis and Discussion

In our experiment, we use Chi-Square to analyze the relationships between the player enjoyment towards each run and their perceived improvement towards each run.

$$x^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

Both of the two variables come from the questionnaire. In this analysis, we categorize the player enjoyment that scales 1 to 5 into 3 categories. An enjoyment level of 1 and 2 means the player does not enjoy this round of play and we categorize it as 0 in the chi-square test. An enjoyment level 3 means the player has no general feelings towards this round of play and we categorize it as 1 in the chi-square test. An enjoyment level of 4 and 5 means the player enjoys this round of play and we categorize it as 2 in the chi-square test.

Table 1. Chi-Squared Tests

	Value	df	p
x^2	28.961	2	<0.01
n	219		

From the Chi-Square Tests table, we could see that the chi-square statistic is 28.961 and the p-value is less than 0.001. Therefore, at a significance level of 0.001, we can conclude that the association between the variables is statistically significant.

In a total of 219 sets of data, players would enjoy the gameplay more when they have a perceived enjoyment since there are 32 sets of data that have an enjoyment of 2, which is higher than the expected value of 16.781. The count for enjoyment level of 0 is 19 which is also lower than the expected value of 31.507. The player would also have a lower enjoyment score when they do not have a perceived improvement, since the data for the enjoyment with a perceived improvement of 1, which is no improvement, is skewed towards low enjoyment score compared to the expected value. With a perceived improvement of 1, the count number for the 0 enjoyment is 73 which is higher than the expected count of 60.493 and the count number for the 2 enjoyment is 17 which is lower than the expected count of 32.219. All in all, we could conclude that players who have a perceived improvement would enjoy the game more.

Table 2. Contingency Tables

Perceived improvement (Yes - 0, No - 1)		Enjoyment 0	Enjoyment 1	Enjoyment 2	Total
0	Count	19.000	24.000	32.000	75.000
	Expected count	31.507	26.712	16.781	75.000
1	Count	73.000	54.000	17.000	144.000
	Expected count	60.493	51.288	32.219	144.000
Total	Count	92.000	78.000	49.000	219.000
	Expected count	92.000	78.000	49.000	219.000

5. Conclusion

The purpose of this study was to investigate the impacts of perceived improvement on players' enjoyment. As expected, the hypothesis was fully supported. Players who perceived improvement felt more enjoyed during the gameplay. More importantly, players who did not perceive improvement felt significantly less enjoyed. It appears that players are expecting skill improvement during the gameplay, and view this element as one of the indispensable parts of the game. Thus, If the game meets the player's expectations for improvement, the player's enjoyment will increase; if not, the player's enjoyment will decline severely.

The key recommendation based on our findings is to ensure that the player's desires are met for each round of the gameplay. A game should be designed to let the player feel that he is making improvements for each round. We demonstrated the correlation between the perceived improvement and enjoyment, to help illustrate the practices to use when designing games powered by various challenges.

References

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Appendix

Appendix A: post-run survey

1. How good do you think you are at platform or auto-run games?
   - ☐ Very good
   - ☐ Good
   - ☐ Fair
   - ☐ Poor
   - ☐ Very poor

2. How much did you enjoy the game?
   - ☐ Very good
   - ☐ Good
   - ☐ Fair
   - ☐ Poor
   - ☐ Very poor

3. Given the opportunity, how likely are you to play this game again?
   - ☐ Strongly want to play this game
   - ☐ Mildly want to play this game
   - ☐ Have no feelings
   - ☐ Mildly do not want to play this game
   - ☐ Strongly do not want to play this game

4. How much did you feel that you improved?
   - ☐ Very much
   - ☐ A fair amount
   - ☐ Somewhat
   - ☐ A little
   - ☐ Not at all

5. Why did you stop playing the game?

   _____

   _____

Appendix B: mid-run survey

Did you feel you improved at this run?

- ☐ Yes
- ☐ No

Did you feel you improved at this run?

- ☐ Yes
- ☐ No

Did you feel you improved at this run?

- ☐ Yes
- ☐ No

Did you feel you improved at this run?

- ☐ Yes
- ☐ No

Did you feel you improved at this run?

- ☐ Yes
- ☐ No

Did you feel you improved at this run?

- ☐ Yes
- ☐ No

Did you feel you improved at this run?

- ☐ Yes
- ☐ No

Did you feel you improved at this run?

- ☐ Yes
- ☐ No